

1 Power Analysis for Run-In Clinical Trials under AR(1) and
2 General Correlation Structures

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4 August 15, 2026

5 Abstract

6 **Background.** The companion Paper 1 of this compendium derived closed-form power
7 formulas for run-in and common close observations under compound symmetry. Com-
8 pound symmetry assumes all pairs of observations are equally correlated regardless of
9 temporal separation, an assumption that is empirically rejected in most longitudinal-trial
10 data. $AR(1)$, $Corr(Y_{ijk}, Y_{ijk'}) = \rho^{|k-k'|}$, is the natural alternative. **Methods.** We derive
11 the variance of the treatment-effect estimator under $AR(1)$ via two routes: (i) the Wood-
12 bury identity applied to the $AR(1)$ marginal covariance, and (ii) a moment-based approach
13 using the cross-covariance between run-in and treatment-period means. The two routes
14 target different estimands: the GLS variance of the slope-difference effect and the variance
15 of the simple phase-mean difference. The GLS derivation accommodates arbitrary J_0 run-
16 in observations and arbitrary J_2 common close observations. **Results.** A taxonomy of
17 five correlation cases is presented, and change-score correlations are derived in closed form
18 under compound symmetry and $AR(1)$. Numerical comparisons using parameters from
19 Alzheimer's disease neuroimaging trials show that, when the two structures are matched
20 at a common nominal correlation parameter ρ , the $AR(1)$ sample-size estimate is smaller
21 than the compound-symmetric estimate (by 4-36% at $\rho = 0.5$), with the deviation growing
22 as ρ increases. A narrow anticonservative regime exists at large J_0 and small ρ , where
23 compound symmetry understates the variance by at most about 6%. **Conclusions.** At a
24 matched nominal ρ , the compound-symmetry assumption underlying Paper 1 is generally
25 conservative: it overstates the required sample size, and increasingly so as the autocorre-
26 lation grows. The exception is a narrow regime with many run-in observations and low
27 autocorrelation, where it is anticonservative by at most about 6%. The $AR(1)$ formulas
28 derived here quantify both effects and should be used directly when serial correlation is
29 expected; this conclusion is specific to the common-nominal- ρ matching convention stated
30 in the text.

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54 **1 Introduction**

55 The efficiency of run-in designs depends on the correlation structure of the re-
56 peated measurements. Under compound symmetry, the correlation between any
57 two change scores is constant, and the run-in observations contribute information
58 about the random slope b_i through a simple averaging mechanism. Under AR(1),
59 the correlation decays with temporal separation, and the information content of
60 run-in observations depends on how far they are from the treatment period.

61 Paper 1¹ assumed $R = \sigma^2(I_J + \mathbf{1}_J\mathbf{1}_J')$, which arises from the change-score parame-
62 terization under independent errors.² extended the run-in framework to two-period
63 linear mixed models but retained the compound symmetry assumption. We extend
64 here to AR(1) residual correlation, where $\text{Corr}(e_{ij}, e_{ij'}) = \rho^{|j-j'|}$ for $0 < \rho < 1$.

65 ³ studied optimal designs for autocorrelated longitudinal data but did not consider
66 the run-in setting.⁴ and⁵ provide general treatments of correlation structures in
67 longitudinal models.⁶ proposed a damped-exponential family of correlation struc-
68 tures with $\text{Corr}(e_{ij}, e_{ij'}) = \rho^{|j-j'|^\theta}$, which contains compound symmetry ($\theta = 0$)
69 and AR(1) ($\theta = 1$) as the two boundary members; the comparison developed
70 here is thus a comparison of the two ends of that family, and the intermediate

members offer a route to relaxing the binary choice.⁷ addressed the effect of intra-individual variation on change-score analyses, a concern that is amplified under AR(1) correlation.

The `longpower` R package⁸ is the CRAN reference implementation for longitudinal-trial power calculations. Its `power.mmr.ar1()` routine implements the analytic AR(1) sample-size formula of⁹ for mixed models for repeated measures, and its slope-based closed-form routines (`edland.linear.power()`, `diggle.linear.power()`) assume random intercepts and slopes with independent or compound-symmetric residuals. None of these routines accommodates run-in or common close observations under a random-slope model. The closed-form AR(1) derivation developed here supplies that case. As $\rho \rightarrow 0$ (no serial correlation), the AR(1) change-score covariance collapses to the compound-symmetric structure of Paper 1, so the present formulas contain the Paper 1 results as a special case.

2 Correlation structures for longitudinal trials

2.1 Taxonomy of cases

We consider five correlation structures for the raw outcomes Y_{ijk} :

1. **Compound symmetry:** $\text{Corr}(Y_{ijk}, Y_{ijk'}) = \rho$ for all $k \neq k'$. This is equivalent to the random intercept model and produces the $R = \sigma^2(I + \mathbf{1}\mathbf{1}')$ structure in change scores.
2. **AR(1):** $\text{Corr}(Y_{ijk}, Y_{ijk'}) = \rho^{|k' - k|}$. Correlation decays geometrically with lag.
3. **Compound symmetry with variable run-in:** Case 1 with $J_{0,i}$ varying across participants.
4. **AR(1) with variable run-in and common close:** Case 2 with $J_{0,i}$ and $J_{2,i}$ varying.
5. **General (unstructured):** $\text{Corr}(Y_{ijk}, Y_{ijk'}) = \sigma_{|k - k'|}^2 / \sigma^2$.

Throughout, the design quantities J_0 , J_1 , and J_2 correspond to the arguments `r`, `k`, and `f` of the implementing `runinpower` functions.

2.2 Change-score correlations under each structure

Under the consecutive-difference parameterization ($C_j = Y_j - Y_{j-1}$, successive visits separated by t), the correlation between change scores depends on the raw-outcome correlation structure. Here $a = \sigma^2$ and $b = \sigma_b^2$. (The GLS machinery of the next section uses change scores from baseline; the two parameterizations are related by a nonsingular linear map, so the GLS variance is unaffected.)

For the run-in design with $J_0 = 2$, $J_1 = 1$, with visits at times $-2t, -t, 0, t$ and consecutive differences C_1, C_2, C_3 :

Compound symmetry (independent raw errors; the shared intercept component

105 **cancels in differencing):**

$$106 \quad \text{Corr}(C_1, C_3) = \frac{bt^2}{2a + bt^2} \quad (1)$$

$$107 \quad \text{Corr}(C_1, C_2) = \frac{-a + bt^2}{2a + bt^2} \quad (2)$$

108 **AR(1):** for consecutive differences at lag $m = |j - l| \geq 1$,

$$109 \quad \text{Var}(C_j) = bt^2 + 2a(1 - \rho) \quad (3)$$

$$110 \quad \text{Cov}(C_j, C_l) = bt^2 - a\rho^{m-1}(1 - \rho)^2 \quad (4)$$

111 so that

$$112 \quad \text{Corr}(C_1, C_3) = \frac{bt^2 - a\rho(1 - \rho)^2}{bt^2 + 2a(1 - \rho)} \quad (5)$$

$$113 \quad \text{Corr}(C_1, C_2) = \frac{bt^2 - a(1 - \rho)^2}{bt^2 + 2a(1 - \rho)} \quad (6)$$

114 These expressions hold for arbitrary (J_0, J_1, J_2) because they depend only on the lag m ; at
115 $\rho = 0$ they reduce to the compound-symmetry formulas above.

116 **3 Variance under AR(1): GLS approach**

117 **3.1 Marginal covariance**

118 Under AR(1) with residual variance σ^2 and autocorrelation ρ , the raw-outcome covariance is
119 $\text{Cov}(e_{ij}, e_{ij'}) = \sigma^2 \rho^{|j-j'|}$. The change-score residual covariance becomes:

$$120 \quad R_{jl} = \text{Cov}(e_{ij} - e_{i0}, e_{il} - e_{i0}) = \sigma^2(\rho^{|j-l|} - \rho^{|j|} - \rho^{|l|} + 1) \quad (7)$$

121 for $j, l \neq 0$.

122 This R no longer has the rank-one-plus-identity structure, so the Sherman-
123 Morrison formula does not apply. However, the Woodbury identity still reduces
124 Σ^{-1} to a scalar inversion because G remains scalar.

125 **3.2 Woodbury computation**

126 The computation proceeds identically to Paper 1:

$$127 \quad \Sigma^{-1} = R^{-1} - R^{-1}Z(G^{-1} + Z'R^{-1}Z)^{-1}Z'R^{-1} \quad (8)$$

128 with $H = (G^{-1} + Z'R^{-1}Z)^{-1}$ still scalar. The difference is that R^{-1} must be com-
129 puted numerically (or via the AR(1) tridiagonal inverse) rather than by Sherman-
130 Morrison.

Table 1: Per-pair $\text{Var}(\hat{\gamma})$ under AR(1) for varying ρ ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

J_0	$\rho = 0$	$\rho = 0.3$	$\rho = 0.5$	$\rho = 0.7$	$\rho = 0.9$
0	3.0000	2.9100	2.7500	2.5100	2.1900
2	2.1569	2.0132	1.6662	1.1157	0.3963
4	1.3268	1.3902	1.2218	0.8508	0.3036

4 Variance under AR(1): moment-based approach

4.1 Cross-covariance between phase means

An alternative to the GLS approach computes the variance of the pre-post test statistic directly from the moments of the phase means, following the summary-measures approach of¹⁰, specialized here to AR(1) correlation. Let $\bar{X}_0 = \frac{1}{J_0} \sum_{i=1}^{J_0} Y_{i,\text{run-in}}$ and $\bar{X}_K = \frac{1}{J_1} \sum_{j=1}^{J_1} Y_{j,\text{trt}}$ denote the run-in and treatment-period means.

Under AR(1) with parameter $c = \rho$, the cross-covariance is:

$$\text{Cov}(\bar{X}_0, \bar{X}_K) = \frac{\sigma^2}{J_0 J_1} \rho c^{K-1} \frac{(1 - c^{J_0})(1 - c^{J_1})}{(1 - c)^2} \quad (9)$$

where $c = \rho$ (the symbol c here denotes the autocorrelation and is unrelated to the reparameterization $c = \sigma^2 - \sigma_b^2 t^2$ used in Paper 1) and K is the time-index lag from the last run-in visit to the first treatment visit on the visit grid $d = (-J_0, \dots, -1, 1, \dots, J_1)$. Because baseline (time 0) is not itself a phase observation, adjacent phases have $K = 2$. The division by $J_0 J_1$ reflects that $\bar{X}_0$ and $\bar{X}_K$ are means of J_0 and J_1 observations respectively; this matches the implementation in `ar1_cov_phase_means()` below.

4.2 Variance of the change score

The variance of the pre-post difference is:

$$\text{Var}(X_K - X_0) = \text{Var}(X_K) + \text{Var}(X_0) - 2 \text{Cov}(X_K, X_0) \quad (10)$$

The variance of each phase mean involves the sum $S_n = \sum_{i=1}^{n-1} (n-i)c^{i-1}$, which has the closed form:

$$S_n = \frac{n(1-c) - (1-c^n)}{(1-c)^2} \quad (11)$$

Moment-based vs GLS variance (r=2, k=2, rho=0.5):

Moment-based phase-mean difference: 1.2188

GLS slope difference (sigma_b2=0): 0.6364

GLS slope difference (sigma_b2=1): 1.6662

(Different estimands; see next section.)

5 Relationship between the two approaches

The GLS approach (Section 3) and the moment-based approach (Section 4) compute different quantities. The GLS variance is the minimum-variance bound under the assumed model, while the moment-based variance is the variance of the simple averaged change-score estimator $\bar{d}_1 - \bar{d}_0$.

The two approaches do not coincide even when $\sigma_b^2 = 0$. At that boundary the GLS estimator still weights the change scores through the AR(1) residual covariance and estimates the slope-difference parameter γ , whereas the moment-based quantity is the variance of the unweighted phase-mean difference $\bar{X}_K - \bar{X}_0$. With $J_0 = 2$, $J_1 = 2$, $\rho = 0.5$, $\sigma^2 = 1$, and $\sigma_b^2 = 0$, the GLS variance is 0.636 while the moment-based variance is 1.219. When $\sigma_b^2 > 0$ the GLS estimator additionally exploits the random-slope structure of $\Sigma = R + ZGZ'$.

The practical value of the moment-based computation is that it requires no specification of σ_b^2 and gives the exact variance of the simple pre-post difference that an analyst might report without fitting a mixed model. It is not a bound on the GLS variance: at the parameters above the moment-based value (1.219) lies between the GLS variance at $\sigma_b^2 = 0$ (0.636) and at $\sigma_b^2 = 1$ (1.666). Because the two quantities are variances of different estimands, their ratio describes the cost of abandoning the mixed model rather than an efficiency ordering within it. The relative-efficiency question is taken up in Paper 4.

6 Robustness to correlation misspecification

The practical question is whether a trial designed under compound symmetry loses substantial efficiency when the true correlation is AR(1), and vice versa. Throughout this section the two structures are matched at a common nominal correlation parameter ρ ; the comparison, and its direction, depend on this matching convention.

7 Stratified variance for unequal visit patterns

When participants have different numbers of run-in ($J_{0,i}$) or common close ($J_{2,i}$) observations, the total variance is a weighted sum across strata:

$$\frac{1}{n^2} \sum_{s=1}^S n_s \text{Var}(X_K - X_0) \big|_{J_0=J_{0,s}, J_2=J_{2,s}} \quad (12)$$

8 Numerical examples

8.1 Comparison across correlation structures

At the matched nominal $\rho = 0.5$, the AR(1) sample sizes are smaller than the compound-symmetric ones at every J_0 , by 3.6% ($J_0 = 0$) to 36.4% ($J_0 = 1$). Positive serial correla-

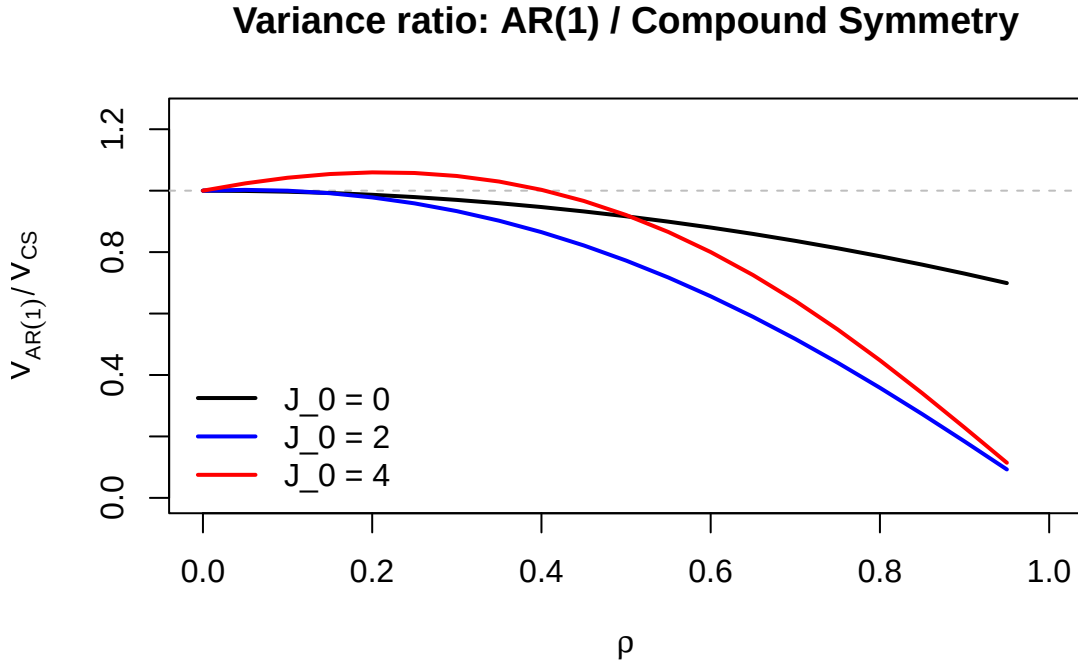


Figure 1: Ratio of AR(1) to compound-symmetry variance at a matched nominal ρ for three run-in configurations ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$). The ratio remains near or below one except at $J_0 = 4$ and small ρ , where it peaks near 1.06.

Table 2: Sample size per group (MIRIAD parameters, $\Delta = 0.25$, 90% power) under compound symmetry vs AR(1).

J_0	J_1	J_2	N (CS)	N (AR(1), $\rho = 0.5$)
0	2	0	385	371
1	2	0	247	157
2	2	0	145	107
3	2	0	100	85
4	2	0	78	72

tion reduces the residual variance of each change score ($R_{jj} = 2\sigma^2(1 - \rho^{|j|}) < 2\sigma^2$), so the compound-symmetric calculation, which corresponds to independent raw errors, is conservative here.

8.2 Sensitivity to rho

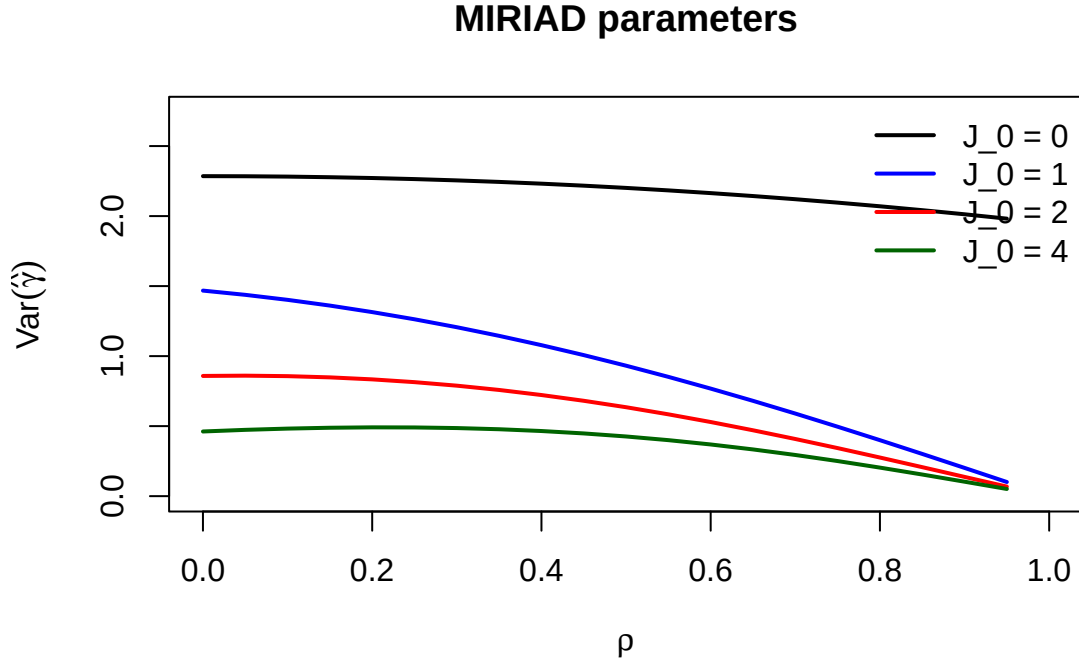


Figure 2: Per-pair variance as a function of ρ under AR(1) for MIRIAD parameters ($\sigma^2 = 0.3363$, $\sigma_b^2 = 0.9745$, $J_1 = 2$, $J_2 = 0$).

9 Discussion

The central practical question is when the compound-symmetry formulas of Paper 1 can be used in place of the AR(1) formulas derived here. The numerical comparisons indicate that the answer is governed by the same slope-to-noise ratio $\sigma_b^2 t^2 / \sigma^2$ that governs the value of run-in observations themselves. When this ratio is large, the random slope term ZGZ' dominates the change-score covariance $\Sigma = R + ZGZ'$, the residual term R contributes little, and the choice between an AR(1) and a compound-symmetric R is immaterial. When the ratio is small, the residual term dominates and the correlation structure becomes the leading determinant of efficiency.

The direction of the error is important for trial design. When the two structures are matched at a common nominal ρ , positive serial correlation reduces the residual variance of every change score ($R_{jj} = 2\sigma^2(1 - \rho^{|j|})$), so the AR(1) variance lies below the compound-symmetric value for most configurations and the compound-symmetric sample size is conservative. In the MIRIAD example at $\rho = 0.5$ the AR(1) sample sizes are 4 to 36 percent smaller than the compound-symmetric

ones, and the gap widens as ρ grows. The exception is a narrow regime with many run-in observations and low autocorrelation ($J_0 = 4$, $\rho \approx 0.2$), where correlation decay makes distant run-in observations less informative than compound symmetry credits them to be; there the compound-symmetric formula understates the variance, by at most about 6 percent in the configurations examined. Both statements are specific to the common-nominal- ρ matching convention; matching the structures on, for example, a common lag-one change-score correlation would alter the comparison.

For applied use we recommend the following. At a matched nominal ρ , the compound-symmetry formula of Paper 1 is conservative for most designs and can serve as a convenient default; the AR(1) formula then refines the sample size downward. When the design places many run-in observations before randomization and the expected autocorrelation is low, the AR(1) formula should be used directly, because the compound-symmetric formula can understate the variance by up to about 6 percent in that regime. When the correlation structure cannot be pinned down from prior data, evaluating the AR(1) formula over a plausible range of ρ and taking the largest resulting variance yields a safe sample size. The moment-based computation of Section 4 addresses the simple pre-post estimator and is not a bound on the GLS variance.

10 Conflict of interest

The author declares no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The `runinpower` R package implementing the AR(1) variance formulas (`covariance-ar1.R`), validation tests, and the shared bibliography across the four-paper compendium are openly available in the project repository. Specific paths are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x. Principal packages used are the in-package `runinpower` source (the AR(1) machinery is in `covariance-ar1.R`), `tinytest` (testing harness), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. The AR(1) variance formulas are deterministic given fixed input parameters; no random-number generation is used anywhere in this manuscript, and re-running the code chunks reproduces every table and figure exactly.

Data and code availability. The R package source is at R/. The companion manuscripts (Papers 1, 3, 4) are at analysis/report/{01,03,04}-*/. The manuscript source and shared bibliography (references.bib, a symlink to ../../references.bib) are at analysis/report/02-correlation/.

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